Article Analysis: Understanding DNA Replication

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In the article, “Recent advances in understanding DNA replication: cell type–specific adaptation of the DNA replication program”, discusses with its audience how the development of specific cells are perfectly articulated to fit their host in therapies, or procedures requiring cell growth. Embryonic development is an increasingly large study that requires cell division research to properly occur. DNA, being the main cells toggled with during embryonic stem cell research, are needed to meet specific needs of the cells they aim to assist (Aze, 2018). This experiment explores differences between differentiated stem cells and embryonic stem cells. The two cells are very different in reproduction, growth rates, and regulation. The goal being to make DNA synthesis a more flawless procedure, researchers study each individual cell to obtain the best methods of production (Aze, 2018). Similarly to a factory in which workers produce goods at a rapid, yet efficient rate, the researchers aim to do the same with DNA synthesis.

The embryonic cells have proved to grow and replicate very rapidly, whereas the differentiated cells reduce DNA synthesis production and tend to have shorter livelihood. The adaptation of working with the embryonic stem cells provides many benefits in certain aspects of DNA replication, such as being able to provide more stability when their DNA is transferred. Both versions of DNA synthesis have their benefits, and it is important to keep testing each type of cell to produce the best results for patients in need. A large reason for this study is due to the many types of cell-killing cancers and how researchers would be able to work more effectively to solve a large majority of our population's health. With cancer rates increasing, studies like these are essential to prolong the lives of innocent patients who are suffering every day. While there are no “cures” to cancer that are 100% fool-proof, studying embryonic stem cells and their replication rates will prove immense value for the medical research world.

References

Aze, A., Maiorano, D., Limas, J. C., & Cook, J. G. (2019). Preparing for DNA replication: Origin licensing and the choice to proliferate or remain quiescent. *Nature Reviews Molecular Cell Biology, 20*(11), 603–620. https://doi.org/10.1038/s41580-019-0156-4